

RESEARCH INTERESTS

Lattice gauge theory and mathematical physics; probability and statistical mechanics; random matrix theory, including Weingarten calculus and β -ensembles.

PUBLICATIONS AND PREPRINTS

1. **Deconfinement for $SO(3)$ Lattice Yang–Mills at Strong Coupling.** Preprint, 2026. arXiv:2605.16162.
2. **Formalization of QFT: Osterwalder–Schrader Axioms for the Free Field in Lean 4** (with Michael R. Douglas, Sarah Hoback, and Anna Mei). 3rd AI for Math Workshop at the 43rd International Conference on Machine Learning (ICML), Seoul, South Korea, July 2026. Workshop paper and poster. arXiv:2603.15770.
3. **$U(N)$ lattice Yang–Mills in the 't Hooft regime.** Submitted, 2025. arXiv:2510.22788.
4. **Dynamical approach to area law for lattice Yang–Mills** (with Sky Cao and Scott Sheffield). To appear in *Proceedings of the American Mathematical Society*. Preprint, 2025. arXiv:2509.04688.
5. **Expanded regimes of area law for lattice Yang–Mills theories** (with Sky Cao and Scott Sheffield). Under revision for *Annals of Probability*. Preprint, 2025. arXiv:2505.16585.
6. **Sharp estimates for large N Weingarten functions.** To appear in *Letters in Mathematical Physics*. Preprint, 2025. arXiv:2502.15892.
7. **On the Limit of the Tridiagonal Model for β -Dyson Brownian Motion** (with Alan Edelman and Sungwoo Jeong). Submitted, 2024. arXiv:2411.01633.
8. **Geometric Derivation of the Finite N Master Loop Equation** (with Omar Abdelghani). Preprint, 2023. arXiv:2309.07399.
9. **Edgeworth-type expansion for the one-point distribution of the KPZ fixed point with a large height at a prior location** (with Ruixuan Zhang). *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, 62(2):1455–1477, 2026. arXiv:2210.04999.

SEMINAR AND CONFERENCE TALKS

1. *Expanded Regime of Area Law for Lattice Yang–Mills Theories.* **International Congress of Mathematicians**, July 2026.
2. *Confinement and Deconfinement in Lattice Yang–Mills Theory.* **Peking Summer in Probability**, June 2026.
3. *Area Law and Mass Gap for Lattice Yang–Mills at Strong Coupling.* **MIT Probability Seminar**, May 2026.
4. *From Weingarten Functions to Random Trees and the Brownian CRT.* **2026 Spring Eastern Sectional Meeting of the American Mathematical Society, Boston College**, March 2026.
5. *Area Law and Mass Gap for Lattice Yang–Mills at Strong Coupling.* **Brin Math Research Center: Statistical Mechanics of Lattice Systems**, March 2026.
6. *Lattice Yang–Mills at Strong Coupling.* **Northeast Probability Seminar**, November 2025.
7. *Lattice Yang–Mills at Strong Coupling.* **University of Chicago Probability Seminar**, October 2025.
8. *Lattice Yang–Mills at Strong Coupling.* **UC San Diego Probability Seminar**, September 2025.
9. *On the Limit of the Tridiagonal Model for β -Dyson Brownian Motion.* **Northeast Probability Seminar**, November 2024.
10. *Asymptotics of the one-point distribution of the KPZ fixed point conditioned on a large height at an earlier point.* **MIT Integrable Probability Seminar**, April 2023.

CONFERENCE POSTERS

1. *Formalization of QFT: Osterwalder–Schrader Axioms for the Free Field in Lean 4.* **International Conference on Machine Learning (ICML)**, July 2026.
2. *From Weingarten Functions to Random Trees and the Brownian CRT.* **Random Explorations: From Random Walks to Random Geometry, IMSI**, July 2026.
3. *Area Law and Mass Gap for Lattice Yang–Mills at Strong Coupling.* **Probabilistic Paths to Quantum Field Theory**,

Simons Collaboration Annual Meeting, May 2026.

AWARDS AND DISTINCTIONS

Fall 2024	<i>NSF Graduate Research Fellowship</i>
Fall 2022	<i>MIT Presidential Fellowship</i>
Spring 2022	<i>NYU Hollis Cooley Memorial Prize</i>
Spring 2021	<i>NYU Undergraduate Mathematics Award</i>

EDUCATION

2022–2027	<i>Ph.D. in Mathematics</i> , Massachusetts Institute of Technology Expected May 2027. Advisor: Scott Sheffield.
2018–2022	<i>B.A. in Honors Mathematics</i> , New York University, May 2022

TEACHING AND MENTORSHIP

January 2026	<i>Directed Reading Program</i> . Mentored Diego Caballero and Davit Kldiashvili in a reading course on Berestycki and Powell's <i>Gaussian Free Field and Liouville Quantum Gravity</i> .
Fall 2025	<i>Recitation Leader</i> , 18.02 Multivariable Calculus.
Summer 2024	<i>Summer Program in Undergraduate Research</i> . Mentored Enrique Rivera in the project <i>U(1) lattice gauge theory and random height functions</i> .
Spring 2024	<i>Teaching Assistant/Grader</i> , 18.676 Stochastic Calculus.
January 2024	<i>Directed Reading Program</i> . Mentored Brandon Wang, Enrique Rivera, and Matija Delic in a reading course on Le Gall and Miermont's <i>Scaling Limits of Random Trees and Planar Maps</i> .
Fall 2023	<i>Teaching Assistant/Grader</i> , 18.338 Random Matrix Theory.
Summer–Fall 2023	<i>Undergraduate Research Opportunity</i> . Mentored Omar Abdelghani in the project <i>Geometric Derivation of the Finite N Master Loop Equation</i> .
January 2023	<i>Directed Reading Program</i> . Mentored Saba Lepsveridze in a reading course on Van Handel's <i>High Dimensional Probability</i> .

ACADEMIC SERVICE

Referee for:

- *Annales Henri Poincaré*
- *Bulletin of the American Mathematical Society*
- *Communications in Mathematical Physics*
- *Courant Journal of Pure and Applied Mathematics*
- *Electronic Journal of Probability*